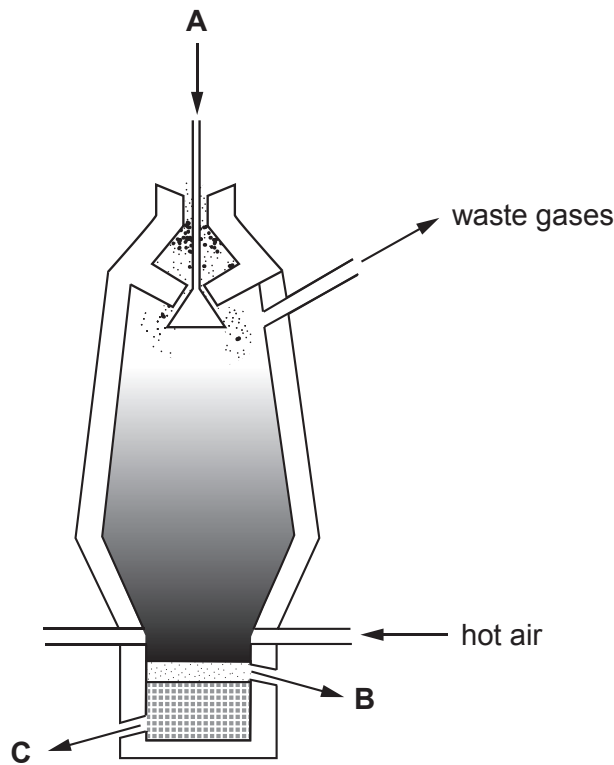


Answer all questions.

1. (a) The diagram shows where substances enter and leave the blast furnace in the extraction of iron from iron ore.



coke	slag	iron	limestone	iron ore
------	------	------	-----------	----------

- (i) Use the substances in the box to complete the following sentences. [3]

The **three** raw materials which enter the furnace at **A** are ,
..... and

Product **B** is and product **C** is



(ii) One reaction that takes place in the furnace is



Underline the element which is removed from the iron(III) oxide during the reaction. [1]

iron

oxygen

carbon

(b) Statements **D**, **E** and **F** show the three steps needed to prepare a sample of copper(II) chloride in the laboratory. The steps are **not** in the correct order.

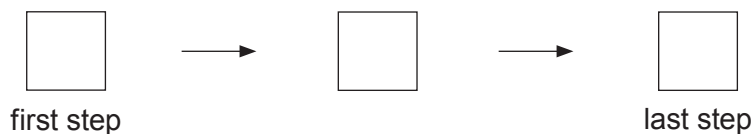
D filter to remove excess copper(II) oxide

E leave the copper(II) chloride solution to evaporate at room temperature

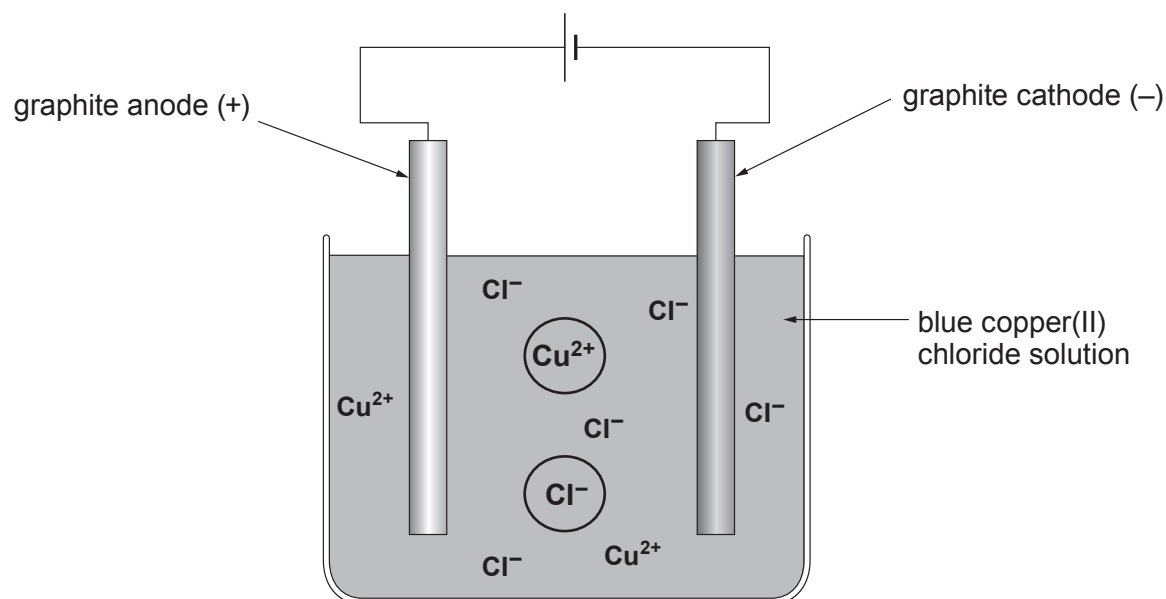
F add excess copper(II) oxide to dilute hydrochloric acid

Complete the flow chart by putting the **letters** in the correct order.

[2]



(c) The diagram shows the apparatus used to obtain copper from copper(II) chloride solution.



electrolyte	electrodes	electricity	electrolysis
-------------	------------	-------------	--------------

- (i) Choose words from the box to complete the following sentences. [2]

The breaking down of copper(II) chloride solution using an electric current is called

.....

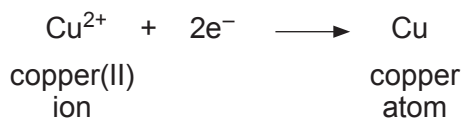
Electric current enters and leaves the copper(II) chloride solution by the

.....

- (ii) **By drawing arrows on the diagram**, show the direction in which the **circled** ions move during the process. [1]



(iii) The reaction occurring at the cathode is:



I. Put a tick (✓) in the box next to what e^{-} represents in the equation. [1]

electricity

☐

electron

☐

ion

☐

atom

☐

II. Copper(II) ions, Cu^{2+} , are removed from the solution during the process.

Put a tick (✓) next to the statement which describes what, if anything, happens to the colour of the copper(II) chloride solution. [1]

the blue solution turns darker

☐

the blue solution turns paler

☐

the blue solution turns yellow

☐

nothing happens to the colour of the blue solution

☐

(iv) Put a tick (✓) next to the gas which is formed at the anode. [1]

oxygen

☐

hydrogen

☐

chlorine

☐
